

PRACTICAL ACTIVITY 1: Encryption and Decryption Using Caesar and Vigenère Ciphers

Objective:

The objective of this activity is to demonstrate the process of encrypting and decrypting messages using classical cryptographic techniques, specifically the Caesar Cipher and the Vigenère Cipher. The activity also aims to verify the results using online encryption tools to ensure accuracy.

Verification Tools Used:

CAESAR CIPHER: <https://www.codbolt.com/tools/caesar-cipher>

VIGENÈRE CIPHER: <https://www.prodevtool.com/tools/vigenere-cipher>

Part 1: Caesar Cipher

1. **Plaintext:** DATA SECURITY IS IMPORTANT

Key Shift: 3

Encryption Process:

PLAINTEXT	CIPHERTEXT	PLAINTEXT	CIPHERTEXT
D	G	I	L
A	D	S	V
T	W	I	L
A	D	M	P
S	V	P	S
E	H	O	R
C	F	R	U
U	X	T	W
R	U	A	D
I	L	N	Q
T	W	T	W
Y	B		

Therefore, the ciphertext is **GDWD VHFXULWB LV LPSRUWDQW**.

Decryption Process: (Shifting the ciphertext three positions backward)

CIPHERTEXT	PLAINTEXT	CIPHERTEXT	PLAINTEXT
G	D	L	I
D	A	V	S
W	T	L	I
D	A	P	M
V	S	S	P
H	E	R	O
F	C	U	R
X	U	W	T
U	R	D	A
L	I	Q	N
W	T	W	T
B	Y		

Therefore, the plaintext is **DATA SECURITY IS IMPORTANT.**

2. *Plaintext:* PROTECT YOUR PASSWORD

Key Shift: 5

Encryption Process:

PLAINTEXT	CIPHERTEXT	PLAINTEXT	CIPHERTEXT
P	U	R	W
R	W	P	U
O	T	A	F
T	Y	S	X
E	J	S	X
C	H	W	B
T	Y	O	T
Y	D	R	W

O	T	D	I
U	Z		

Therefore, the ciphertext is **UWTYJHY DTZW UFXXTWI**.

Decryption Process: (Shifting the ciphertext five positions backward)

CIPHERTEXT	PLAINTEXT	CIPHERTEXT	PLAINTEXT
U	P	W	R
W	R	U	P
T	O	F	A
Y	T	X	S
J	E	X	S
H	C	B	W
Y	T	T	O
D	Y	W	R
T	O	I	D
Z	U		

Therefore, the plaintext is **PROTECT YOUR PASSWORD**.

Part 2: Vigenère Cipher

1. **Plaintext:** CYBER SECURITY IS ESSENTIAL

Keyword: KEY

Key Expansion:

Plaintext: CYBERSECURITYISESSENTIAL

Key : KEYKEYKEYKEYKEYKEYKEYKEYKEYKEY

Encryption Process:

PLAINTEXT	P	KEY	K	P + K (MOD 26)	CIPHERTEXT
C	2	K	10	12	M
Y	24	E	4	2	C
B	1	Y	24	25	Z
E	4	K	10	14	O
R	17	E	4	21	V
S	18	Y	24	16	Q
E	4	K	10	14	O
C	2	E	4	6	G
U	20	Y	24	18	S
R	17	K	10	1	B
I	8	E	4	12	M
T	19	Y	24	17	R
Y	24	K	10	8	I
I	8	E	4	12	M
S	18	Y	24	16	Q
E	4	K	10	14	O
S	18	E	4	22	W

S	18	Y	24	16	Q
E	4	K	10	14	O
N	13	E	4	17	R
T	19	Y	24	17	R
I	8	K	10	18	S
A	0	E	4	4	E
L	11	Y	24	9	J

Therefore, the ciphertext is **MCZOVQOGSBMRIMQOWQORRSEJ**.

Decryption Process:

CIPHERTEXT	C	KEY	K	C - K (MOD 26)	PLAINTEXT
M	12	K	10	2	C
C	2	E	4	24	Y
Z	25	Y	24	1	B
O	14	K	10	4	E
V	21	E	4	17	R
Q	16	Y	24	18	S
O	14	K	10	4	E
G	6	E	4	2	C
S	18	Y	24	20	U
B	1	K	10	17	R
M	12	E	4	8	I
R	17	Y	24	19	T
I	8	K	10	24	Y
M	12	E	4	8	I

Q	16	Y	24	18	S
O	14	K	10	4	E
W	22	E	4	18	S
Q	16	Y	24	18	S
O	14	K	10	4	E
R	17	E	4	13	N
R	17	Y	24	19	T
S	18	K	10	8	I
E	4	E	4	0	A
J	9	Y	24	11	L

Therefore, the plaintext is **CYBERSECURITYISESSENTIAL**.

Reflection:

This activity helped me better understand how classical cryptographic techniques work by allowing me to actually perform the encryption and decryption processes myself. Instead of only learning the concepts theoretically, applying the Caesar and Vigenère ciphers to real examples made it clearer how readable messages can be transformed into encrypted text. Through this process, I realized how substitution-based encryption hides the original message by systematically changing the characters according to specific rules.

While working with the Caesar cipher, I observed how a simple shift in the alphabet can effectively disguise a message at first glance. However, the activity also made its weaknesses very apparent. Because the shift value is fixed throughout the entire message, it becomes relatively easy to break the cipher using simple techniques such as brute-force attacks or frequency analysis. This made me realize that although the Caesar cipher was useful historically, it does not provide strong security by modern standards.

When I applied the Vigenère cipher, I noticed that it was more complex because it uses a keyword to change the shift for each letter in the message. This creates a polyalphabetic substitution system where the same plaintext letter can produce different ciphertext letters depending on the key. From my perspective, this made the encryption appear more secure and less predictable compared to the Caesar cipher. However, I also learned that even this method has vulnerabilities. With modern cryptanalysis techniques, such as analyzing repeating key patterns or using methods like the Kasiski examination, the cipher can still be broken.

Overall, this activity made me appreciate how classical cryptographic methods laid the foundation for modern cybersecurity. By understanding how these early encryption techniques work and how they can be broken, I gained a better perspective on why modern cryptographic algorithms need to be far more complex and resistant to analytical attacks. This experience helped me see the importance of continuously improving encryption methods to ensure secure communication in today's digital world.